

PALMDALE, CA, USA

WMO: 723820

Lat: 34.629N

Lon: 118.084W

Elev: 2524

StdP: 13.40

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23182

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.3	28.0	-1.2	5.7	48.0	2.7	6.9	47.4	32.2	48.2	28.2	48.9	3.6	200	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.6	102.4	65.2	100.1	64.8	98.0	64.0	69.9	93.8	68.2	92.7	66.6	91.8	14.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	93.6	80.1	59.1	82.1	80.5	55.6	72.3	81.3	35.3	93.2	33.8	92.5	32.4	91.7	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.3	25.5	23.4	DB	20.1	107.4	2.9	2.0	18.0	108.8	16.3	110.0	14.7	111.1	12.6	112.6	(4)
(5)			WB	17.0	73.1	2.7	4.5	15.0	76.4	13.4	79.0	11.8	81.5	9.9	84.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	46.5	48.6	54.3	59.3	67.3	76.2	83.1	82.1	75.2	63.8	52.8	45.2		
	DBStd	14.69	5.79	5.98	6.49	7.42	7.63	6.99	4.83	4.63	6.57	6.43	6.78	5.92		
	HDD50	494	138	91	33	11	0	0	0	0	0	2	47	172		
	HDD65	2730	575	459	333	200	67	7	0	0	8	100	366	615		
	CDD50	5218	28	52	168	288	537	786	1025	995	756	429	132	22		
	CDD65	1979	0	0	2	28	139	343	560	530	314	62	1	0		
	CDH74	27842	3	16	154	574	1965	4723	7832	7263	4151	1056	103	2		
	CDH80	15444	0	1	23	175	902	2639	4718	4352	2260	365	9	0		
(14) Wind	WSAvg	9.8	7.2	9.0	10.4	12.3	12.8	12.3	11.0	10.0	8.8	8.6	7.4	7.5		
(15) Precipitation	PrecAvg	7.0	1.3	1.3	1.1	0.4	0.1	0.0	0.0	0.2	0.2	0.2	1.0	1.0		
	PrecMax	15.6	4.7	6.4	5.2	2.5	1.7	0.7	0.5	1.5	2.1	2.0	5.6	5.3		
	PrecMin	1.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.5	1.3	1.6	1.3	0.6	0.3	0.2	0.1	0.4	0.5	0.5	1.2	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.4	76.1	83.8	90.0	98.2	103.9	106.8	105.5	100.4	91.4	81.2	71.1		
		MCWB	50.3	51.2	54.7	57.9	63.7	63.6	67.5	68.4	65.0	59.4	53.2	50.6		
	2%	DB	66.9	71.6	78.7	84.8	93.2	99.6	103.2	101.8	97.7	87.9	77.1	65.7		
		MCWB	47.8	49.3	53.5	55.6	60.9	63.4	65.6	65.4	64.3	58.0	51.7	48.3		
	5%	DB	63.7	66.8	74.7	81.1	89.9	96.8	100.3	99.5	95.2	84.1	72.7	62.3		
		MCWB	46.6	48.1	52.1	54.5	59.6	62.3	65.0	65.0	62.3	56.3	50.5	46.2		
	10%	DB	59.7	62.5	70.4	76.2	85.6	93.2	97.9	97.1	92.1	80.6	68.0	57.8		
		MCWB	45.3	46.5	50.8	52.8	57.5	61.4	64.3	63.8	61.7	54.9	48.8	45.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	54.6	57.6	60.3	66.4	69.6	72.5	72.4	71.2	64.0	56.7	55.0		
		MCDB	63.4	64.1	75.8	82.9	90.3	93.8	96.0	96.1	91.3	78.1	72.2	64.1		
	2%	WB	51.5	51.7	55.4	57.8	63.1	66.0	69.9	69.4	68.1	60.7	54.1	52.1		
		MCDB	60.3	65.6	73.0	79.2	90.0	93.8	94.0	95.1	89.0	80.2	70.1	60.3		
	5%	WB	49.6	50.0	53.7	55.8	60.5	64.1	68.3	67.6	66.1	58.7	52.2	49.5		
		MCDB	58.1	63.1	70.6	76.5	85.9	91.9	93.0	93.9	87.5	78.2	67.7	57.2		
	10%	WB	47.5	48.4	51.9	54.0	58.8	62.4	66.7	65.8	64.0	56.7	50.6	47.0		
		MCDB	57.3	60.0	67.8	73.0	82.5	89.4	92.0	93.0	85.4	75.7	65.2	55.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.1	23.9	25.9	26.6	27.2	29.3	28.6	29.6	30.3	28.8	27.7	24.4		
		MCDBR	31.7	32.7	33.5	33.9	33.9	33.3	31.7	32.0	32.8	34.4	34.1	30.8		
		MCWBR	17.0	16.0	15.0	14.2	13.4	12.5	11.9	12.9	12.8	14.7	16.3	17.0		
	5% WB	MCDBR	23.2	26.1	29.3	29.9	31.8	31.2	27.4	29.4	28.8	28.9	28.2	23.3		
		MCWBR	13.6	13.6	14.1	13.4	13.1	12.3	11.8	13.4	12.0	12.5	14.2	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.296	0.314	0.335	0.346	0.345	0.365	0.347	0.334	0.321	0.288	0.284		
	taud		2.521	2.510	2.447	2.392	2.388	2.401	2.393	2.420	2.453	2.468	2.543	2.548		
	Ebn at Noon		296	304	305	301	297	296	289	293	294	290	292	290		
	Edh at Noon		26	29	34	37	38	38	38	36	33	30	25	24		
(44) All-Sky Solar Radiation	RadAvg		992	1278	1740	2141	2408	2623	2496	2348	1995	1508	1128	904		
	RadStd		75	137	95	97	115	116	105	62	47	84	49	65		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+362	N/A
(47) Regional (46 neighbors)		+1.13	N/A	-3.20	+1.19	N/A	N/A	N/A	-207	+401	+191

Nomenclature: See separate page